

Summer Topology Program 2026

Topological Data Analysis (TDA)

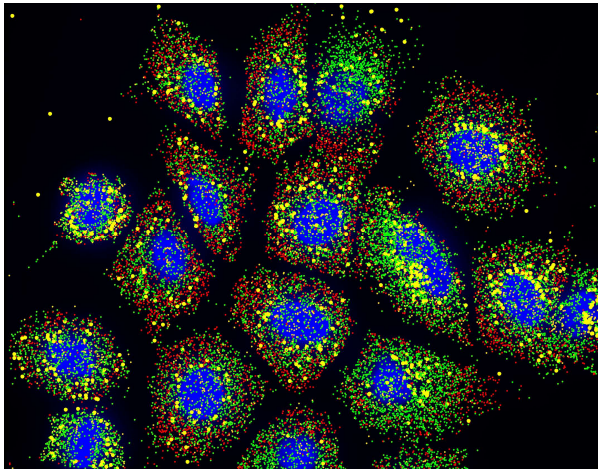
A brief introduction with examples

Presenting: Astrid A. Olave H.

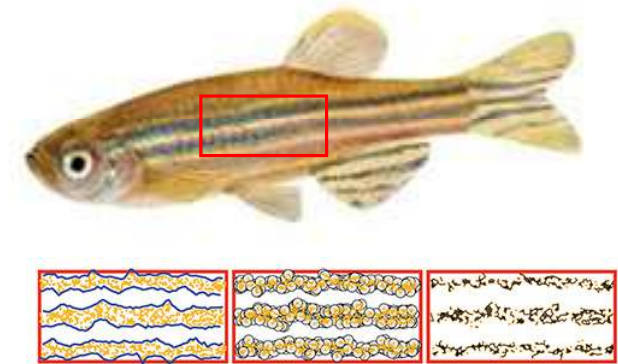


Data has shape and shape is data

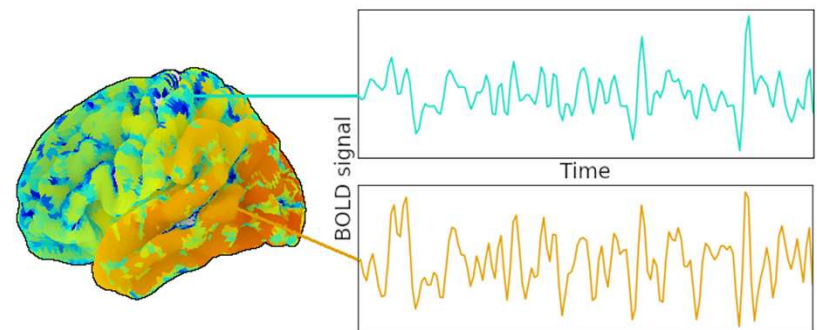
Characterizing RNA
patterns
using persistence



Modelling skin
dynamics in zebrafishes
using persistence



Revealing brain
dynamics
using mapper
algorithm



Topological data analysis aims to find the shape of the data

Topology and geometry offer powerful tools for uncovering the “shape of data”.

So, TDA provides a collection of mathematical, statistical and algorithmic methods design to explore, infer and analyze data trough a topological and geometrical lens, revealing qualitative and sometimes quantitative information about the structure of data.